Preliminary Data Collection

DAEN 459 Fall 2025

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Purpose of Preliminary Data Collection

- Clarifies data availability, structure, and feasibility.
- Guides pipeline architecture and tool selection.
- Supports validation of objectives and stakeholder needs.
- Enables early discovery of quality or access constraints.

Importance of Planning

- Early planning prevents major architectural redesign later.
- Reduces risk of data or knowledge loss between terms.
- Ensures consistent technical workflow and team cohesion.

Types of Preliminary Data

- Internal databases, system logs, cloud storage.
- APIs, web scraping, open-source datasets.
- Pilot data ingestion samples for architecture testing.
- Stakeholder-provided documents and business logic.

Early Data Profiling & Engineering Considerations

- Examine volume, velocity, schema, and update frequency.
- Identify ingestion approach: batch vs. streaming.
- Assess compatibility with proposed ETL/ELT tools.
- Determine computational/storage requirements.

Planning for Consistency Across Semesters

- Set expectations for handoff between semesters.
- Establish coding practices, version control, and documentation standards.
- Get started building reproducibility through notebooks, scripts, and data schemas.
- Ensure project continuity even if team changes (or expands).

Data Quality & Risk Control

- Define validation rules and track nulls, outliers, schema drift.
- Implement lightweight test ingestion pipeline or proof of concept (PoC) scripts.
- Early risk identification:
  access permissions, data latency, API limits.
- Use documentation and contingency planning to anticipate challenges.

Security & Compliance

- Secure data handling policies.
- Encryption strategy (in transit & at rest).
- Role-based access control and credential management.
- Institutional and regulatory compliance: FERPA, GDPR, etc.

Minimum End-of-Semester Deliverables

- Initial roadmap for Spring implementation.
- Preliminary dataset(s) successfully accessed and profiled.
- Drafted architecture diagram (pipeline, data flows, storage layers).
- Prototype ingestion script or test ETL.
- Documentation: data dictionary, access protocol, early results.

Semester Transition Timeline

Phase	Planning (Fall)	Implementation (Spring)
Data Collection	Exploratory	Full-scale ingestion
Architecture	Draft & feasibility	Final build and optimization
Testing	Pilot scripts	Production pipeline validation
Documentation	Planning report	Final deliverables

Preliminary Findings & Validation

- Key insights from pilot ingestion or schema profiling (sample pipeline executed on small dataset).
- Initial performance statistics and data issues identified early (nulls, missing fields, performance).
- Evidence supporting or challenging feasibility assumptions. (Compatibility checks with storage formats (e.g., Parquet, CSV))
- Findings informing infrastructure choices and adjustments recommended for Spring implementation.

Success Indicators for Next Semester

- Scope defined, feasible, and approved by stakeholders.
- No major unknowns related to data access or structure.
- Architecture decisions informed by preliminary tests.
- Team aligned on implementation roadmap.

Summary

- Preliminary data collection drives informed design decisions.
- Consistent planning reduces rework and accelerates development.
- Strong documentation ensures continuity into next semester.
- Strategic alignment now leads to smoother implementation phase.

Next Steps & Discussion

- Finalize architecture outline before semester ends.
- Confirm any outstanding assumptions with stakeholders.
- Identify main (*"top 3"*) implementation tasks for early Spring.